

DESIGN AND DEVELOPMENT OF A PORTABLE UNIVERSAL TESTING MACHINE

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Chapter 1

Introduction

Material testing is a fundamental aspect of engineering and product development, offering critical data on how materials and structural components behave under various loading conditions. Properties such as strength, stiffness, ductility, and elasticity are commonly evaluated using Universal Testing Machines (UTMs), which perform tests like tension, compression, and flexural loading. However, traditional UTMs are typically large, stationary, and expensive, making them less accessible for educational settings, on-site testing, and small-scale prototype evaluation.

Recognising this gap, this project sets out to develop a compact and portable Universal Testing Machine specifically designed for low-load applications up to 1kN. The objective is to create a machine that is not only lightweight and cost-effective but also structurally robust, modular, and easy to operate. Such a device would be ideal for academic labs, engineering competitions, and basic material characterisation tasks, where full-scale UTMs are impractical.

To meet these goals, the design incorporates NEMA 23 stepper motors and a lead screw mechanism for controlled load application, all housed within a rigid aluminium frame to maintain strength while reducing weight. An S-type load cell is used to measure force precisely, and an Arduino-based control system allows integration with a digital interface for ease of monitoring.

Prior to fabrication, the complete model was developed and simulated using Autodesk Fusion 360. Structural analyses were conducted to validate the design under load, ensuring the final prototype will perform reliably in real-world testing scenarios.

Chapter 2

Research Motivation and Objectives

Universal Testing Machines (UTMs) are critical tools in engineering, widely used to evaluate the mechanical properties of materials under tension, compression, bending, and shear. However, conventional UTMs are often bulky, heavy, expensive, and stationary, making them impractical for flexible academic use or on-site testing. Their size and immobility create a barrier for educational environments that require adaptability, particularly in settings where hands-on experimentation and mobility are key.

The motivation for this project stems from the need to make material testing more accessible, interactive, and portable for students and small-scale users. Our goal is to develop a compact and lightweight UTM that can be easily transported and operated in various environments—classrooms, laboratories, workshops, or even competition venues. For instance, in student design contests such as bridge-building challenges, full-scale UTMs are not feasible to move. A portable alternative enables real-time, on-site testing, enhancing both the learning experience and the practicality of project validation.

This project does not aim to replace industrial-grade testing systems but instead focuses on educational support and small-scale prototype evaluation. The intention is to provide students with a hands-on understanding of how testing works, using a machine that is affordable, reliable, and easy to use.

Design Phase Objectives:

- Design a Universal Testing Machine with a 1kN capacity, suitable for testing small material samples or student-built models.
- Ensure portability by optimising the design for minimal size and weight, without compromising structural integrity.
- Develop a detailed 3D CAD model that is ready for simulation and structural analysis.
- Perform virtual testing using simulation tools to validate the design through stress, deflection, and load behaviour analysis.
- Finalise a design that is functionally reliable and ready for fabrication in the next phase.

This design phase lays the groundwork for building and testing the physical prototype, forming a critical step in transforming the concept into a working, educational tool.

Chapter 3

Conceptual Design

The conceptual design of the Portable Universal Testing Machine (UTM) was developed to build a compact and easily transportable device for testing small-scale structural specimens and materials. The design focuses on simplicity, portability, and structural safety, while ensuring it can handle axial loads up to 1kN.

3.1 Frame

The frame forms the primary structure of the UTM (Figure 3) and is designed using aluminium profiles to keep the system lightweight and portable. The upper part of the frame is built using 30×30 mm aluminium square profiles, which provide enough strength and stability to support the moving components (Figure 2). The base uses 20×20 mm aluminium profiles to reduce weight while offering sufficient support (Figure 1). A steel sheet of $700 \text{ mm} \times 300 \text{ mm}$ and 1 mm thickness is mounted on the base to serve as a rigid and flat platform for placing specimens.

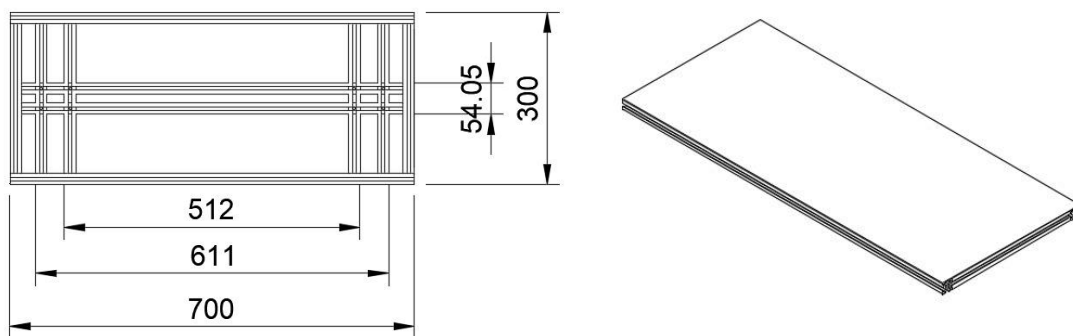


Figure 1 Base of the frame without a steel sheet and with a steel sheet

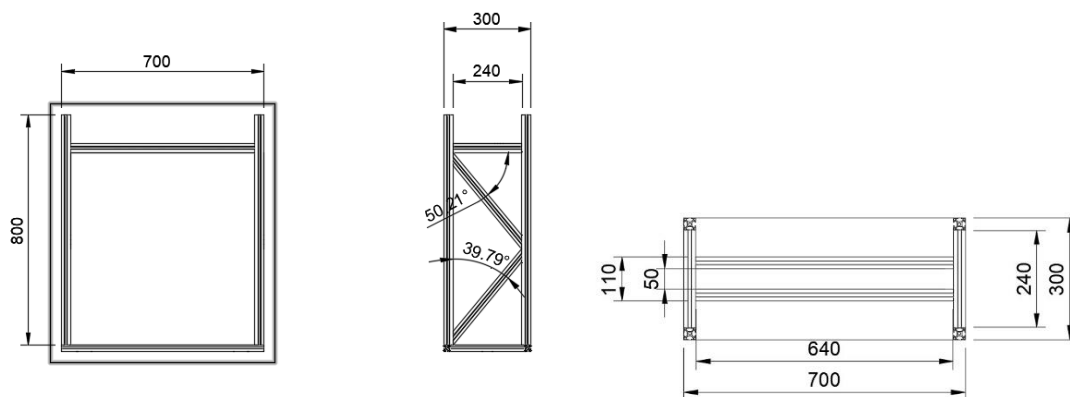


Figure 2 Front View, side view, top view of the frame

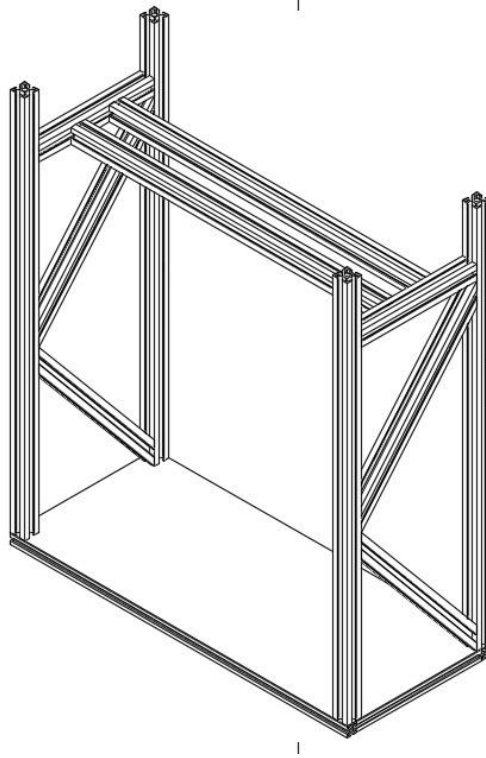


Figure 3 Frame

3.2 Lead Screw Mechanism:

Vertical movement in the system is achieved using two ball screws of 20 mm diameter (Figure 5), positioned vertically on either side of the structure. These are connected to NEMA 23 stepper motors via shaft couplers (Figure 4), allowing the motors to transfer rotational motion to the screws. As the screws rotate, they cause the crosshead to move vertically. Thrust bearings are provided at the lower ends of the ball screws to prevent the base from getting damaged by screw. This setup ensures smooth motion and protects the frame from wear.



Figure 4 Lead screw mechanism

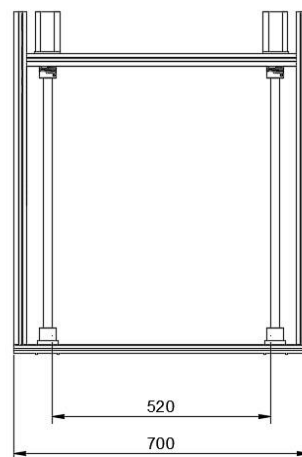


Figure 5 Lead screw mechanism connected with frame

3.3 Movable Bar, Load Cell, and Load Applier

The load is applied through a moving part called the movable bar, which is connected to a load cell and a load applicator (as per Figure 6). The bar is made from an 80×80 mm aluminium profile, explicitly chosen to ensure it does not bend or deform when force is applied during testing. It has nuts fixed inside it, which are engaged by the ball screws. When the ball screws rotate, the nuts move along the screw threads, causing the movable bar to move up or down. This vertical motion is used to apply a load on the specimen. The S-type load cell is mounted below the movable bar to measure the applied force accurately. The load applicator is mounted below the load cell, directly interacting with the specimen and evenly transferring the load. This setup ensures smooth movement, accurate force application, and safe testing conditions.

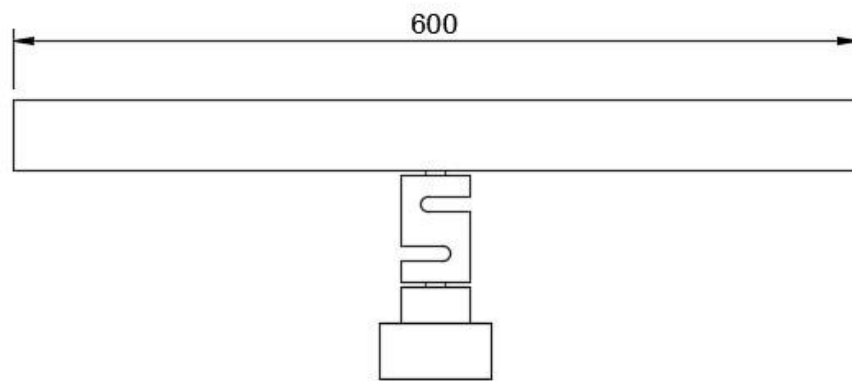


Figure 6 Movable bar along with load cell and load applier

3.4 Final Design:

Therefore, the final design satisfies all the requirements for a portable universal testing machine (UTM). The design looks as follows (Figure 7).

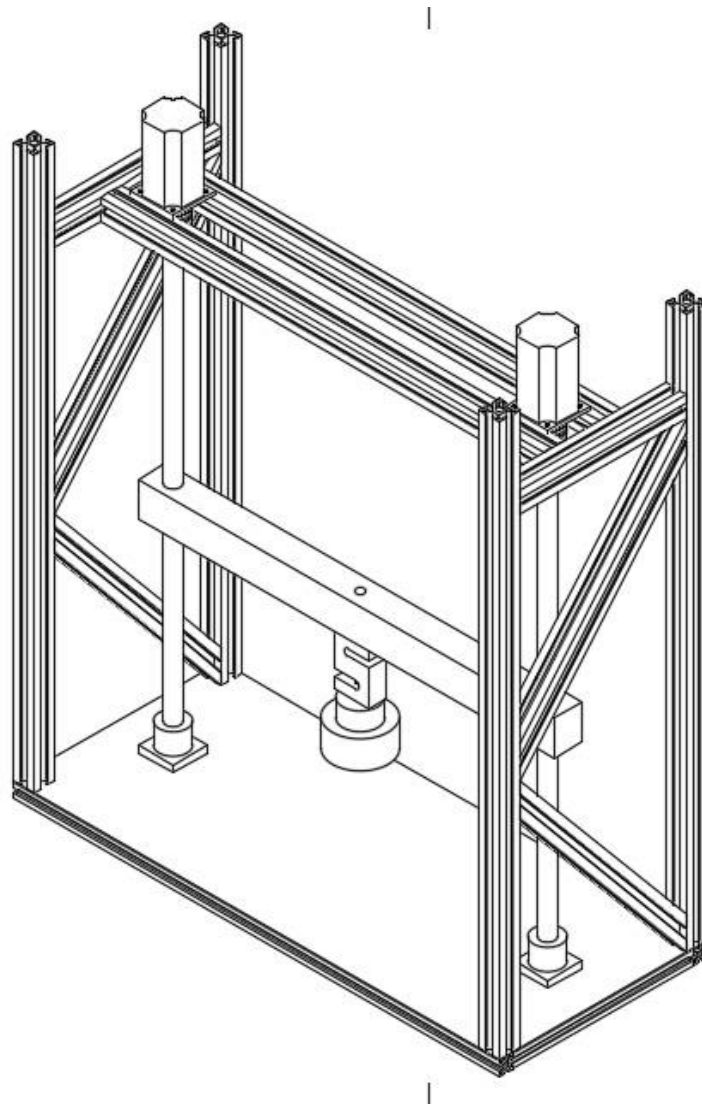


Figure 7 Final Design of UTM

Chapter 4

Structural Analysis and Simulation

As part of the proposed design, we conducted a structural simulation of the Universal Testing Machine (UTM) using Autodesk Fusion 360 to assess its structural integrity and predict its performance under expected loading conditions. The primary goal of the simulation was to ensure that the UTM could reliably sustain the intended load capacity of 1kN while maintaining stability and safety during operation.

The process began with defining the overall dimensions of the UTM and identifying the essential components required for its construction. These included aluminium extrusions with cross-sectional dimensions of 30 mm × 30 mm and 20 mm × 20 mm, along with key mechanical elements such as stepper motors, ball screws, a steel base plate, and other structural parts. Each component was meticulously modelled in Fusion 360 to match real-world dimensions and configurations. Material properties were assigned accurately, aluminium for the frame elements and mild steel for critical load-bearing parts such as the base plate and lead screws (Table 1).

Once all components were modelled, they were assembled using appropriate constraints and mechanical joints to replicate realistic assembly behaviour. This comprehensive model allowed us to simulate load conditions and analyse critical responses such as stress distribution, deformation, and reaction forces. The simulation environment closely reflected the physical design, allowing for a reliable performance evaluation before fabrication.

Table 1 Properties of material used (Aluminium 6061-T6 and Mild Steel)

Properties	Aluminium (6061 T6)	Mild Steel
Density	$2.7 \times 10^{-6} \text{ kg/mm}^3$	$7.85 \times 10^{-6} \text{ kg/mm}^3$
Young's Modulus	69GPa	210GPa
Poisson's Ratio	0.33	0.29 – 0.30
Yield Strength	240 MPa	250 MPa
Ultimate tensile Strength	290 - 310 MPa	400–550 MPa
Thermal Conductivity	167 W/m·K	43–60 W/m·K
Thermal Expansion Coefficient	$23.6 \times 10^{-6} /\text{K}$	$11\text{--}15 \times 10^{-6} /\text{K}$
Specific Heat	896 J/kg·K	490 J/kg·K

The simulation was performed using Fusion 360's integrated analytical tools, which assess the Universal Testing Machine's (UTM) structural behaviour using finite element analysis (FEA) methods. This research was crucial to guarantee the machine's overall stability and safety and to comprehend how various structural elements would react to the given load.

Applying suitable boundary conditions, material characteristics, and a 1kN load to the model was all part of the procedure. Critical performance metrics like von Mises stress, displacement, and deformation were monitored across the entire structure through Fusion 360's simulation module. These findings indicate how the structure would react to actual loading scenarios.

By monitoring these parameters, we could pinpoint areas of potential concern—such as regions of higher stress concentration or excessive deflection—and assess whether any components required design modifications or structural reinforcement. This approach ensured that the final design would be efficient and safe before fabrication.

Chapter 5

Results and discussions of the simulations

The simulation results obtained from Fusion 360 offered critical insights into the structural performance of the portable Universal Testing Machine (UTM) under expected loading conditions. The primary focus of the analysis was to evaluate stress distribution, displacement, and the structure's overall stability when subjected to a load of 1kN.

According to the simulation, the maximum von Mises stress developed in the UTM was approximately 42 MPa (shown in Figure 8), significantly lower than the yield strength of aluminium 6061-T6, the primary material used for the frame (240 MPa). This substantial margin indicates that the structure remains within safe operational limits under the applied load.

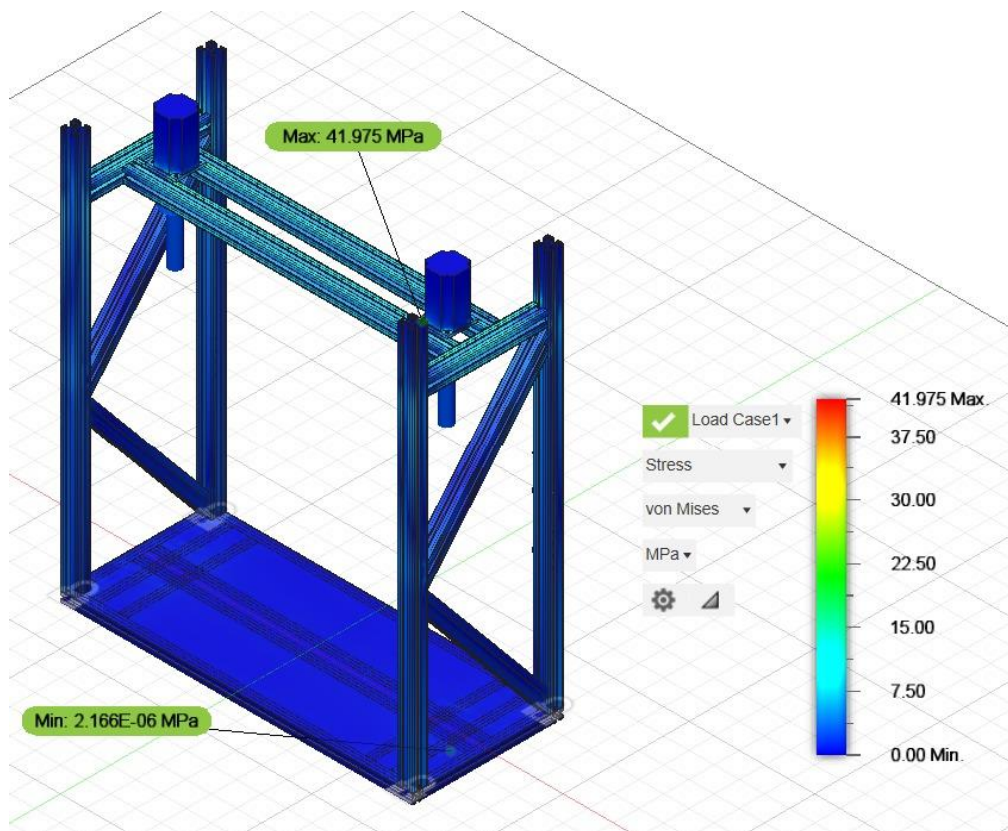


Figure 8 Von Mises stress simulation of the frame of UTM

The simulation also provided detailed insights into the internal force distribution within the structural components of the UTM. The vertical aluminium columns were found to experience axial forces of approximately 250 N, while the long horizontal beams carried slightly higher loads, with the maximum reaching around 280 N. These horizontal bars bear the most significant load among all the structural members in the aluminium frame.

Additionally, the simulation revealed the maximum deflection occurring at the mid-span of the central horizontal beam, where the load is applied. This deflection, measured at 0.50 mm, is primarily attributed to the force being distributed between two lead screws, which use the load symmetrically but still induce bending in the central region. Despite this deflection, the values remain within acceptable limits, confirming that the structure can maintain its integrity and functionality under the expected operating conditions.

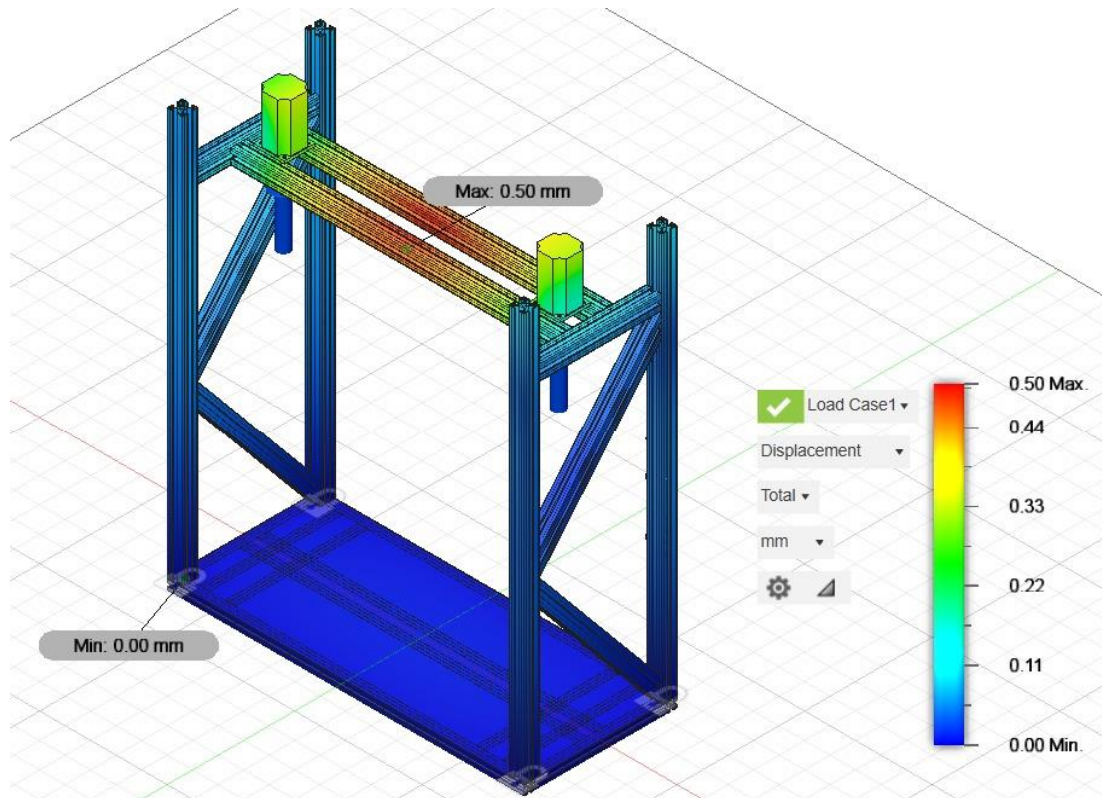


Figure 9 Deflection occurred during the simulation of the frame of UTM.

A separate structural analysis was conducted for the frame and the lead screw mechanism of the Universal Testing Machine (UTM), as we encountered challenges in simulating the entire assembly together within Fusion 360. From the analysis, the maximum von Mises stress observed was approximately 25 MPa (refer to Figure 10), which is significantly lower than the yield strength of both aluminium and mild steel, the primary materials used in the design.

The maximum deflection recorded was around 0.113 mm (refer to Figure 11), occurring near the shaft coupler region. This value is minimal and indicates that the system experiences very limited deformation even under high load conditions. These results confirm that the structural components of the UTM—including the lead screw assembly—can maintain stability and structural integrity during operation, thereby ensuring safe and reliable performance under the specified load capacity.

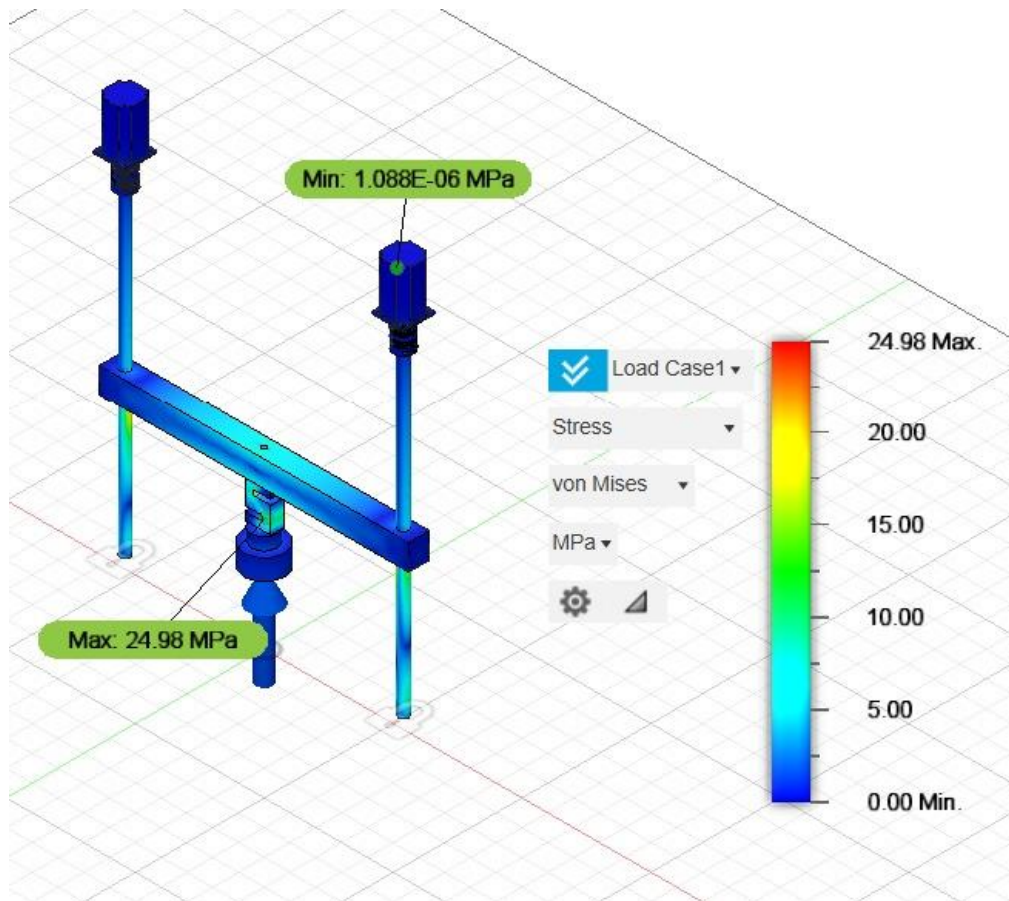


Figure 10 Von mises stress in lead screw part of UTM

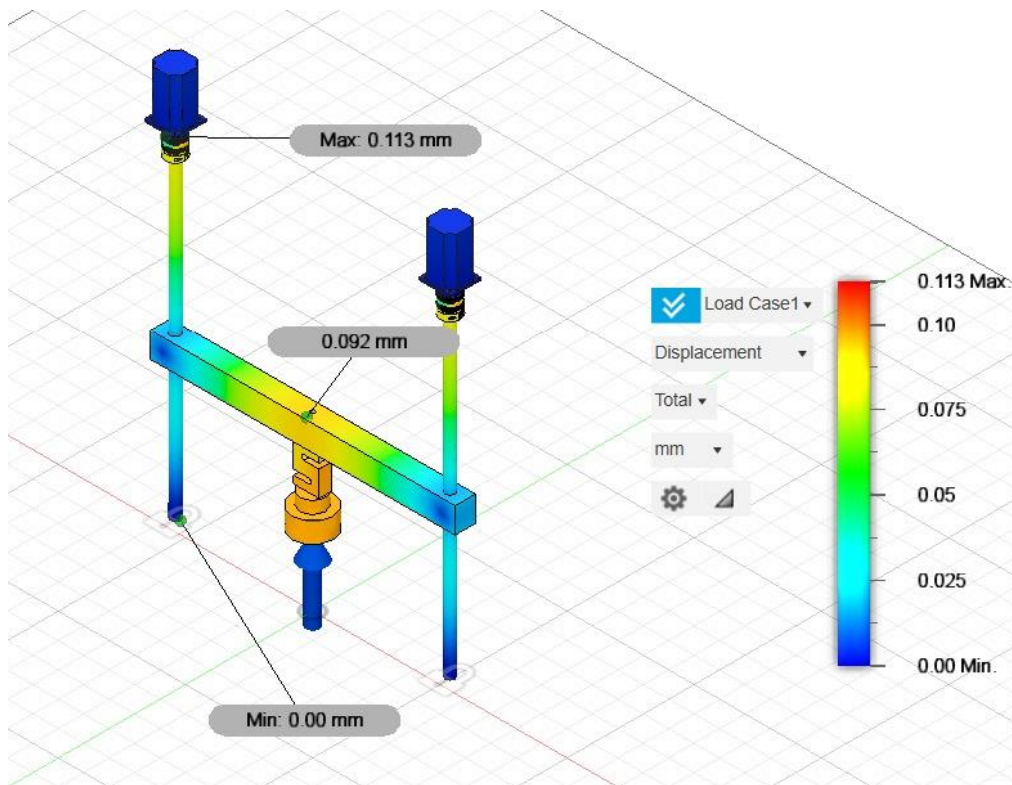


Figure 11 Deflection occurred in lead screw part of UTM

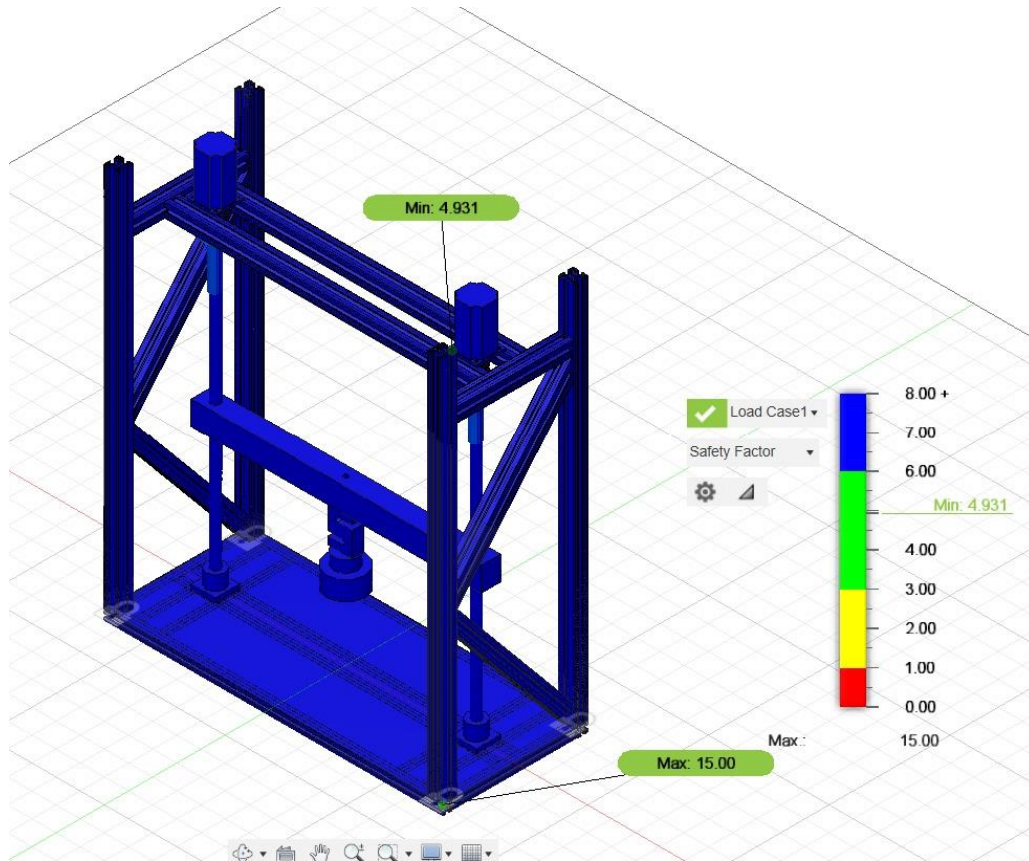


Figure 12 Safety Factor of UTM

Furthermore, the simulation calculated a minimum safety factor of 4.931 (Figure 12), indicating that the structure operates well within its material limits under the applied loading conditions. This high safety factor demonstrates that the design meets the strength requirements and provides a significant margin for unexpected loading or variations in material properties. Combined with the previously observed stress levels and deflection, this result confirms the structural reliability and safety of the proposed UTM design for its intended 1kN load capacity.

Chapter 6

Future Work

The design phase of the UTM was completed, and the transition from the modelling stage to the manufacturing phase began, where the 3D model would be turned into a fully functional physical prototype. Based on the finalised CAD model design, the immediate focus shifted to carefully selecting and acquiring all the necessary materials, mechanical components, and electronic modules. To ensure a practical and efficient build, material selection was based on factors such as cost-effectiveness, durability, and weight constraints.

Fabrication will begin as soon as all required parts are sourced. The process will involve assembling the structural frame, accurately aligning the load-bearing components, and integrating mechanical connections to ensure precise load application and result measurement. In parallel, the team will work on the electrical part using an Arduino Uno microcontroller along with an Analogue-to-Digital Converter (ADC) module. This configuration enables precise control of the NEMA 23 stepper motors, allowing for specific loading conditions.

The testing strategy adopted was based on a displacement-controlled method, where an S-type load cell measured the corresponding loads after known displacement values were applied. Although an LVDT (Linear Variable Differential Transformer) sensor was considered for measuring displacement as an output for a given load, the team found that integrating it increased the complexity of mounting and design, as well as the overall system weight. Given the UTM's portable nature, the team decided to continue with the displacement-controlled technique in this project.

Chapter 7

Conclusion

The portable UTM was designed to be lightweight, cost-effective, and structurally stable, making it suitable for small-scale and academic-level material testing. Through 3D CAD modelling and structural simulations conducted in Autodesk Fusion 360, the team validated that the design could safely withstand loads up to 1kN, with a minimum safety factor of 4.93, ensuring robust performance under expected loading conditions.

The system's integrated key components are NEMA 23 stepper motors, dual lead screws, an S-type load cell, and an aluminium frame, enabling automated load application and accurate force measurement. Simulation results revealed a maximum von Mises stress of 42 MPa, which is below the material's yield strength.

Although the UTM was not intended to replace industrial-scale machines, its compactness, portability, and affordability made it an efficient solution for educational laboratories and prototype testing. As the team moved into the fabrication phase, the objective was to construct a fully functional prototype demonstrating a real-world application. The project successfully delivered a design that can balance functionality and ease of use, showing its strong potential for low-load material testing in accessible settings.

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